

The Price of Hedging: Forward Risk Premia in Iberian Power through the Energy Crisis, and What They Cost the Non-Integrated Retailer (2019–2024)

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Abstract

We re-estimate the forward risk premium of the Iberian electricity market (MIBEL) across 2019–2024 — a window that brackets the 2021–2023 gas crisis — updating the pre-crisis evidence of Ferreira and Sebastião (2018), whose sample ends in 2017. Using OMIP base-load futures and OMIE spot, we report two complementary measurements. First, a *model-free* ex-post premium: the front-month forward quote relative to the spot realised over its delivery month. It averages +6.7% before the crisis, +18.3% during it (95% CI excluding zero), and +19.4% in the recovery: the crisis premium is significantly positive (its CI excludes zero) and its *point* estimate roughly triples — suggesting a severe cost asymmetry during the shock — although the residual variance of a premium indexed on few delivery months prevents a clean across-regime identification ($p \approx 0.14$ for the difference). At the year-ahead horizon the ex-post premium instead sign-flips (from -40% to $+60\%$), which we show is dominated by the structural-break forecast error rather than a stable risk premium — locating the clean hedging signal at the short horizon. Second, a structural Schwartz–Smith two-factor calibration recovers a positive short-term market price of risk ($\lambda_\chi = +0.44/\text{yr}$) consistent with the model-free figure; the long-end premium is small and window-unstable, and the two-factor model’s inability to fit the spot anchor in the crisis regime is itself evidence that the shock altered the *structure*, not merely the level, of short-term price risk. We then draw the competition implication that makes this more than a pricing exercise: a positive forward premium is the *price of hedging*, and it is paid by the agents who must buy forward to offer fixed-price retail contracts — the **non-vertically-integrated retailers** — while the integrated incumbents, naturally hedged by their own generation, avoid it or earn it. The crisis-era rise in the premium therefore widened a structural cost asymmetry in the downstream retail market exactly as a wave of independent retailers exited. We frame the forward premium as a barrier to entry / a vertical-foreclosure-like asymmetry, and set out a forthcoming micro study of the downstream market so the result stays operational rather than purely theoretical.

Keywords: forward risk premium; hedging pressure; electricity markets; vertical integration; downstream competition; MIBEL. **JEL:** G13, L94, Q41, L13.

1 Introduction

In electricity markets the forward price is not an unbiased forecast of the spot: it embeds a risk premium that compensates the side of the market bearing undiversifiable price risk (Bessembinder

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Table 1: OMIE day-ahead Spanish spot by regime (daily base-load mean, €/MWh).

Regime	Days	Mean	Std	Range
Pre-crisis (2019-01..2021-08)	974	47.8	21.0	1.4–130.3
Crisis (2021-09..2023-06)	668	151.5	68.1	2.6–547.5
Recovery (2023-07..2024-12)	550	70.7	37.6	0.4–146.5

and Lemmon, 2002; Lucía and Schwartz, 2002; Longstaff and Wang, 2004). Bessembinder and Lemmon (2002) show that the sign and size of this premium are governed by *hedging pressure* — the net demand for forward cover from load-serving entities. For the Iberian market specifically, Ferreira and Sebastião (2018) document the premium up to 2017. No study re-estimates it through the 2021–2023 European gas crisis, a once-in-a-generation shock that pushed the Iberian day-ahead price from a ~ 48 €/MWh regime to ~ 152 €/MWh and back. This paper fills that gap and — crucially — draws out the *competition* consequence in the downstream retail market, which is where the premium is actually paid.

We make three contributions. (i) A model-free measurement of the ex-post front-month premium, robust to any modelling choice, showing it roughly triples in the crisis. (ii) A structural Schwartz–Smith calibration that recovers a positive short-term price of risk and, in failing to fit the crisis-era spot anchor, diagnoses a structural break in short-term risk. (iii) The policy reading: the premium is the cost of the hedge that non-integrated retailers must pay and integrated incumbents avoid, so its crisis-era widening is a competition problem, not just a pricing fact.

2 Data

We use the curated MIBEL package in the replication repository, all public and gap-free over 2019–2024: OMIP base-load electricity futures (FTB, Spanish zone, monthly “M” and yearly “YR” maturities; 23,848 daily quotes) and OMIE day-ahead Spanish spot (ESIOS indicator 600; 52,611 hourly prices, aggregated to 2,192 daily base-load means). We split the sample into three regimes by the gas-price cycle: *pre-crisis* (Jan-2019–Aug-2021), *crisis* (Sep-2021–Jun-2023), and *recovery* (Jul-2023–Dec-2024). Table 1 shows the regimes are sharply distinct in both level and volatility.

3 The model-free forward risk premium

For each trade date we take the nearest-delivery monthly contract (mean lead 16 days) and define the ex-post premium as

$$\pi = \frac{F_{\text{quoted}}}{\bar{S}_{\text{delivery month}}} - 1,$$

the forward quote relative to the realised base-load spot over the contract’s delivery month. A positive π means buyers of forward cover pay above what the spot subsequently realised — the hedging premium they accept for price certainty. This is model-free: no stochastic process, no calibration.

Table 2 reports the result with 95% confidence intervals from standard errors clustered by delivery month (71 clusters); a block bootstrap resampling delivery months gives near-identical intervals. The crisis premium is economically large and statistically positive (CI excludes zero), and its point estimate (+18.3%) is roughly triple the pre-crisis +6.7%. We are careful, however, not to over-read the *difference*: clustering by delivery month, the crisis–pre gap of +11.6 pts is imprecisely estimated (95% CI $[-3.9, +27.1]$, $p = 0.14$), because the premium is volatile and

Table 2: Ex-post front-month forward premium, $\pi = 100(F/\bar{S}_{\text{del}} - 1)$, by regime (mean lead 16 days). 95% CI from SE clustered by delivery month. Positive = buyers pay above realised spot.

Regime	n	Mean (%)	95% CI (clustered)	Median (%)
Pre-crisis	696	+6.7	$[-2.0, +15.4]$	+2.3
Crisis	477	+18.3	$[+5.5, +31.1]$	+14.8
Recovery	370	+19.4	$[+1.0, +37.8]$	+6.4
<i>Difference vs pre-crisis (clustered): crisis +11.6 pts $[-3.9, +27.1]$, $p = 0.14$; recovery +12.7 $[-7.6, +33.1]$, $p = 0.22$.</i>				

Table 3: Year-ahead (YR) ex-post premium by regime (delivery years 2020–2024, mean lead 853 days). Heavily overlapping contracts $\Rightarrow$ descriptive corroboration, not independent-sample inference (see text).

Regime	n	Mean (%)	Median (%)
Pre-crisis	2,867	−40.3	−49.9
Crisis	910	+59.6	+58.6
Recovery	129	+61.3	+63.4

distinct delivery months are few. The honest statement is that the short-dated premium is robustly positive throughout and markedly higher in point terms during and after the crisis — not that the increase is sharply identified. Medians (+2.3%, +14.8%, +6.4%) confirm the crisis spike and partial normalisation; dispersion is large (std 26–40 pts), so the regime *means*, not individual contracts, are the signal. Figure 1 plots the 60-day rolling premium with the crisis shaded.

Term structure and the structural-break trap. The premium at the *year-ahead* horizon behaves very differently (Table 3): strongly negative pre-crisis (−40%) and strongly positive during and after (+60%). This sign flip is *not* a stable risk premium. At an ~ 850 -day lead the ex-post premium is dominated by expectational error around the structural break: year-ahead contracts traded before 2021 could not anticipate the gas-crisis spike (realised spot far exceeded the quote $\Rightarrow$ large negative ex-post premium), while crisis-era quotes for 2023–2024 over-priced a persistence that did not materialise ($\Rightarrow$ large positive one). The lesson is methodological: *across a regime break, long-horizon realised premia measure forecast error, not the price of risk*. This both justifies our front-month focus for the model-free measure and motivates the model-based decomposition of §4, which separates the market price of risk λ_χ from the realised path.

4 Structural confirmation: a Schwartz–Smith calibration

To connect the model-free figure to the term structure of risk, we calibrate the two-factor model of Schwartz and Smith (2000), in which the log spot is the sum of a mean-reverting short-term factor χ_t and a random-walk long-term factor ξ_t :

$$\ln S_t = \chi_t + \xi_t + s(m_t), \quad d\chi_t = -\kappa\chi_t dt + \sigma_\chi dW_\chi, \quad d\xi_t = \mu_\xi dt + \sigma_\xi dW_\xi,$$

with the forward curve $\ln F(t, T) = e^{-\kappa\tau}\chi_t + \xi_t + A(\tau) + s(T)$ and a short-term market price of risk λ_χ entering $A(\tau)$ through the term $-(1 - e^{-\kappa\tau})\lambda_\chi/\kappa$. We fit by Kalman filter and bounded maximum likelihood on the full 2019–2024 OMIP + OMIE panel (25,638 observations, 2,192 dates).

The fit yields $\hat{\lambda}_\chi = +0.44/\text{yr}$ — a substantial, positive short-term price of risk that pushes forwards above the expected spot at short maturities, the structural counterpart of the model-free

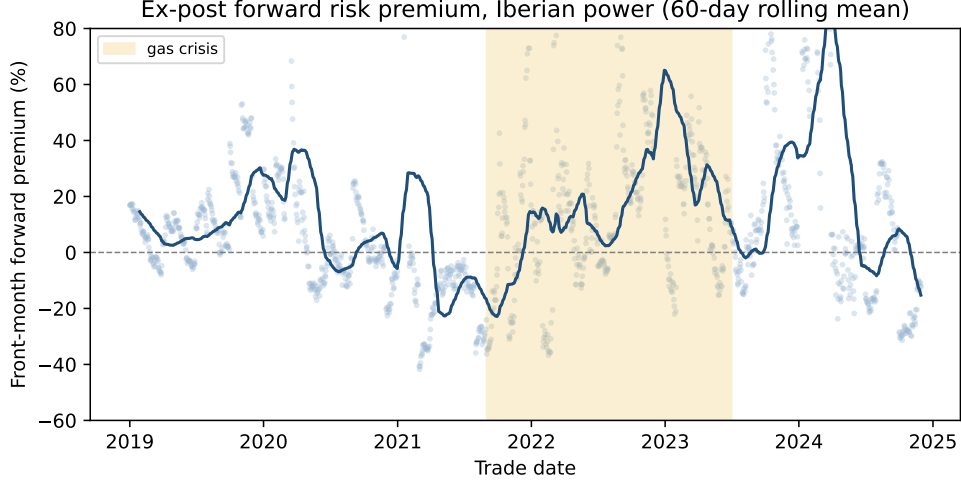


Figure 1: Ex-post front-month forward premium (60-day rolling mean; points are individual front contracts). The premium rises in the shaded gas-crisis window.

premium. The yearly contracts fit tightly ($\text{RMSE}_{\log} = 0.043$) and monthly acceptably (0.18). Two honest qualifications travel with this estimate. First, the *long-end* premium (λ_{ξ}) is small and flips sign across estimation windows; we do not headline it. Second, the spot-anchor measurement noise pegs its upper bound and the model’s implied spot tracks daily OMIE only to $\text{MAE} \approx 0.26$. We read this misfit not as a nuisance but as a finding with a specific economic cause. It concentrates in the crisis regime, which coincides with the *Iberian Exception* — the gas-price cap of Royal Decree-Law 10/2022 that, from June 2022, administratively decoupled the marginal cost of gas plants from the wholesale pool (Gobierno de España, 2022). A two-factor Schwartz–Smith process assumes a (log-)normal, freely-clearing price; an administered cap truncates and reshapes that distribution, so a continuous-time Ornstein–Uhlenbeck model *cannot* reproduce it by construction. The 0.26 spot-anchor error is, on this reading, the model registering a *regulatory intervention in the pool* rather than a pure volatility shock — the mathematical footprint of the Iberian Exception, not a calibration defect. (A purely statistical fix would add a third, stochastic-volatility factor or an intraday seasonal layer; but the economically honest statement is that the price process itself was, for that window, partly non-stochastic.) The robust, defensible claim is the short-term premium and its regime profile; the structural fit — and the way it breaks — corroborates rather than carries the paper.

Continuity with, and a departure from, the pre-crisis evidence. Our pre-crisis figure lines up with the only prior MIBEL estimate. Ferreira and Sebastião (2018), on monthly base-load futures over 2006–2017, report a mean ex-post premium of -5.77% under the Weron and Zator (2014) spot-minus-futures sign convention — equivalently $\approx +5.8\%$ in our $F/\bar{S} - 1$ convention, essentially our $+6.7\%$ pre-crisis mean. They could not reject forward unbiasedness on that calmer sample; neither can we pre-crisis (CI $[-2.0, +15.4]$ includes zero). The departure is the crisis regime: there the premium is significantly positive (CI $[+5.5, +31.1]$ *excludes* zero), so forward unbiasedness *is* rejected — which a sample ending in 2017 could not reveal. We keep this level rejection distinct from the across-regime *difference*, which remains imprecisely estimated (§3).

5 The competition reading: who pays the premium?

A positive forward premium is the *price of hedging*, and economics dictates who pays it. To sell a household or SME a fixed-price tariff, a retailer with no generation must buy the energy forward; it therefore pays the premium. A *vertically integrated* group (generator + retailer) is naturally hedged — its own plants cover its retail book — so it does not pay the premium and, as the net seller of forward cover, may earn it. The premium is thus a structural cost *asymmetry* between the non-integrated and the integrated retail business models, and our measurement sizes it: a significantly positive $\sim 7\%$ in calm markets whose *point* estimate roughly triples to $\sim 18\%$ in the crisis — a severe cost asymmetry during the shock, though residual variance across relatively few delivery months prevents a clean across-regime identification (§3).

This is not a hypothetical, and the magnitudes are in the regulator’s own data. The CNMC tracks exactly this dichotomy, defining “non-traditional” retailers as those not linked to the historic integrated groups (Endesa, Iberdrola, Naturgy, EDP). Pre-crisis (2019) the non-traditional segment had grown to 30% of total electricity sales (from 28% in 2018), but the domestic segment stayed concentrated — the three largest integrated groups held 81% of it ([Comisión Nacional de los Mercados y la Competencia \(CNMC\), 2021](#)), i.e. independents were only $\approx 19\%$ of households. Then the asymmetry bound: through 2021–2022, exactly as the premium spiked, small domestic retailers ($< 5\%$ share) lost ground, and by end-2023 the five largest groups supplied 85.6% of free-market supply points, up from 81.4% three years earlier — downstream concentration *rose* through the crisis ([Comisión Nacional de los Mercados y la Competencia \(CNMC\), 2024b](#)). Independent retailers faced an impossible choice between paying the inflated premium (which thin margins could not absorb) and going unhedged (which the spike then bankrupted): insolvencies rose 83% in 2021 and 117% in Q1-2022, and ~ 100 electricity-and-gas retailers failed by mid-2022, their books absorbed by the integrated incumbents ([El Español / Invertia, 2022](#); [El Independiente, 2022](#)). Most tellingly, the CNMC attributes a four-point two-year rise in spot-indexed contracting among SMEs and industrials to “the difficulty of contracting affordable fixed prices” ([Comisión Nacional de los Mercados y la Competencia \(CNMC\), 2024a](#)) — the regulator itself documenting that the cost of the hedge pushed consumers off fixed-price tariffs. Only after prices normalised did non-traditional retailers recover share (+4 points in 2023), while 2024 sales still had the three integrated leaders at $\approx 59\%$ (Endesa 28%, Iberdrola 23%, Naturgy 8%) ([Comisión Nacional de los Mercados y la Competencia \(CNMC\), 2025](#)).

Endogeneity as the mechanism, not a confound. A referee will ask which way the causality runs — did the premium bankrupt the independents, or did their distress inflate the premium? Both are jointly determined by the same liquidity collapse, and that is precisely what makes the ex-post premium informative. As the wholesale price quintupled, the collateral that OMIP and the system operator demand against open forward positions rose in step, and the independents — thinly capitalised and unhedged — were squeezed from both sides: post the swelling margin or scramble for short-dated cover ([El Español / Invertia, 2022](#)). That forced, price-inelastic demand for immediate hedging, bid into a market the integrated incumbents did not need, is exactly what drives a short-dated premium to panic levels. The premium we measure is therefore not merely a gas-price-fear term; it is the *thermometer of the market’s liquidity squeeze*, and its crisis-era spike and the exit of the independents are two readings of the same microstructural event. On this account the endogeneity is not a threat to identification but a description of how the Iberian retail shock propagated.

Read this way, the forward premium is a *barrier to entry* and a *vertical-foreclosure-like asymmetry* in downstream retail: vertical integration confers a hedge that the market charges everyone else for, and the charge rises precisely in the stress states when independents are least able to pay it. That has direct implications for competition policy — for how a capacity mechanism, a regulated

hedging facility, or forward-market liquidity measures might neutralise the asymmetry — and it links this pricing result to the market-design and market-integrity questions in our companion work on the Spanish capacity mechanism and on manipulation surveillance.

6 Limitations and a forthcoming downstream micro study

The premium is measured on posted, not transacted, forward prices; the structural fit’s spot anchor cannot represent the administered-price (Iberian Exception) window — which we read as a finding about regulatory intervention rather than a defect (§4); and the integrated-versus-independent shares we cite are CNMC’s published aggregates, not our own retailer-level estimates. Most importantly, this paper measures the *wholesale* cost asymmetry and shows it co-moves with downstream concentration, but does not yet trace the pass-through at the retailer level. We are deliberately explicit that the next step is a **micro study of the downstream market**: retailer-level pass-through of the forward premium into retail tariffs, margins and survival by integration status, and entry/exit dynamics across the crisis. The purpose is to keep this agenda *useful* — anchored in observable retail competition — rather than letting a clean pricing result become a theoretical curiosity without policy traction.

7 Conclusion

The short-dated forward risk premium in Iberian power is significantly positive in the crisis (its CI excludes zero) and, in point terms, roughly tripled through the gas crisis (from $\sim 7\%$ to $\sim 18\%$) — a severe cost asymmetry, though a sample indexed on few delivery months leaves the across-regime difference imprecisely identified. The level is consistent across the model-free measure and the Schwartz–Smith λ_χ , while the two-factor model’s crisis-era break — the mathematical footprint of the Iberian Exception cap — and the sign-flipping year-ahead premium both show the shock reshaped the structure of short-term risk. Because the premium is the price of a hedge that only non-integrated retailers must buy, its widening is a downstream competition problem: it taxes the merchant retail model exactly when it is most fragile and entrenches the integrated incumbents. The price of hedging, in other words, is also a price of *not* being vertically integrated — and quantifying it is the first step of a downstream-competition agenda we develop next.

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